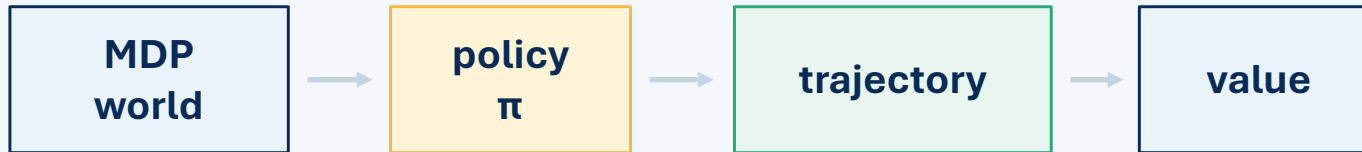


Artificial Intelligence

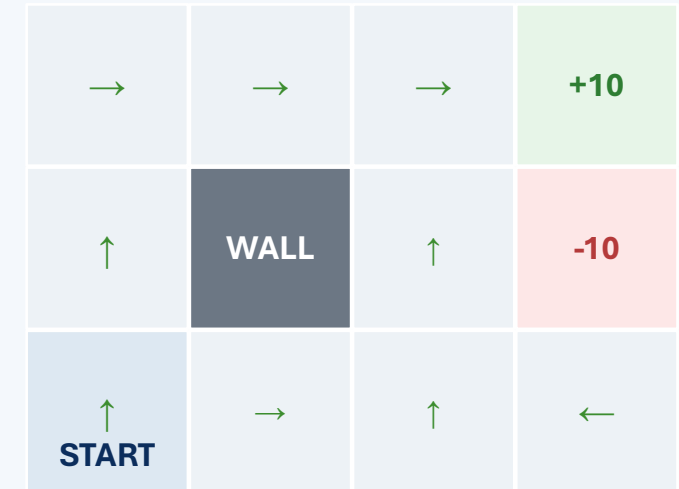
Lesson 12: MDP II

Policies, Return, and Value



An MDP describes what can happen. A policy describes what the agent will do.

Shared Gridworld Lab



Today's mission

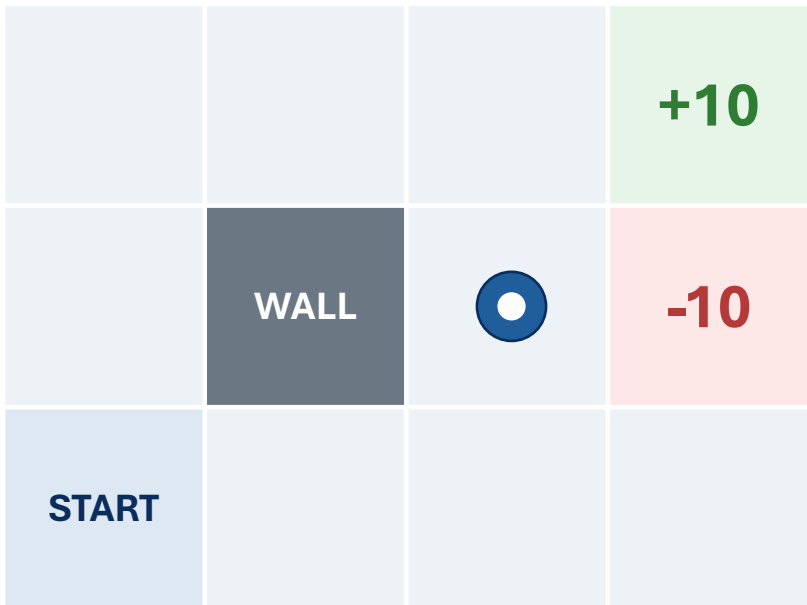
Policy and value give meaning to behavior over time.

- 1 Define policies and value functions.**
- 2 Interpret cumulative reward.**
- 3 Explain what makes a policy optimal.**

Performance target: interpret a policy, compare simple long-term returns, distinguish V^π from Q^π , and justify better behavior.

Retrieval: MDP Foundations

The MDP gives the environment, not the behavior.



Piece	Meaning
S	states
A	actions
T	transitions
R	rewards

Where does this MDP tell us which action the agent should actually take?

It does not.
That is why we need a policy.

The model tells us what can happen—not what to do

A complete environment model still leaves behavior unspecified.

?	?	?	+10
?	WALL	?	-10
?	?	?	?

No policy yet: each state still asks “what action?”

MDP world

- Where the agent can be
- What actions are legal
- What outcomes may happen
- What rewards are received

Missing behavior layer

- Which action to take in each state
- What to do after an unexpected transition
- How to compare long-term consequences

The environment is modeled. The agent still needs a behavior rule.

A policy fills in the decisions

The arrows are the policy.

→	→	→	+10
↑	WALL	↑	-10
↑	→	↑	←

$$\pi(s) = a$$

Read it in words

When the agent is in state s , the policy chooses action a .

Why this matters

If stochastic movement sends the agent somewhere unexpected, the policy still tells it what to do next.

A policy is not one action—or one path

Uncertainty means the agent needs a rule for every state it may reach.



Plan = what I intend to do. Policy = what I will do from each possible state.

Deterministic policy

One selected action for each state.

Each nonterminal state receives one action arrow.

→	→	→	+10
↑	WALL	↑	-10
↑	→	↑	←

$$\pi(s) = a$$

Interpretation

If the agent is in this state, choose this action.

Important

The policy chooses the action;
the transition model determines what may happen next.

Stochastic policy

Sometimes the agent randomizes its choice of action.

			+10
	WALL		-10

Focus state

Policy probability
 $\pi(\text{East} \mid s) = 0.70$
 $\pi(\text{North} \mid s) = 0.30$

Which action the agent chooses



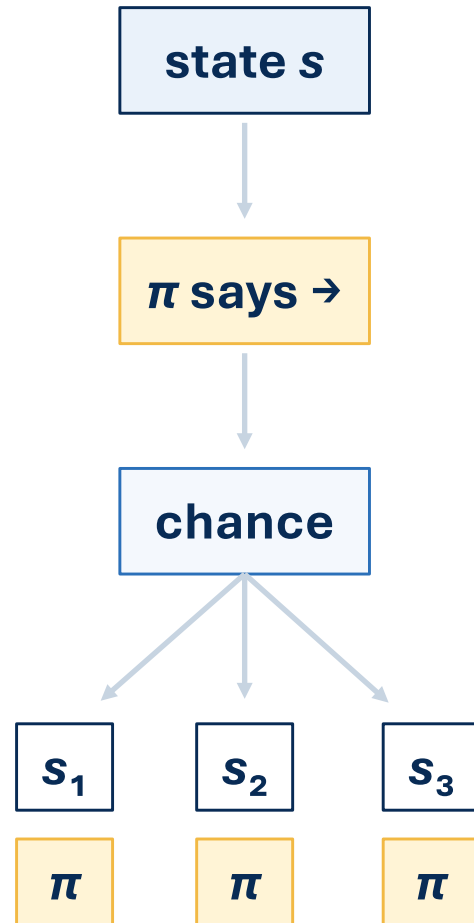
Transition probability
 $P(s' \mid s, \text{East})$

What happens after selecting East

Policy probability = how the agent chooses.
Transition probability = how the environment responds.

Following a policy creates future trajectories

A policy plus stochastic transitions produces many possible paths.



Trajectory

a sequence of states, actions, and rewards that unfolds through time

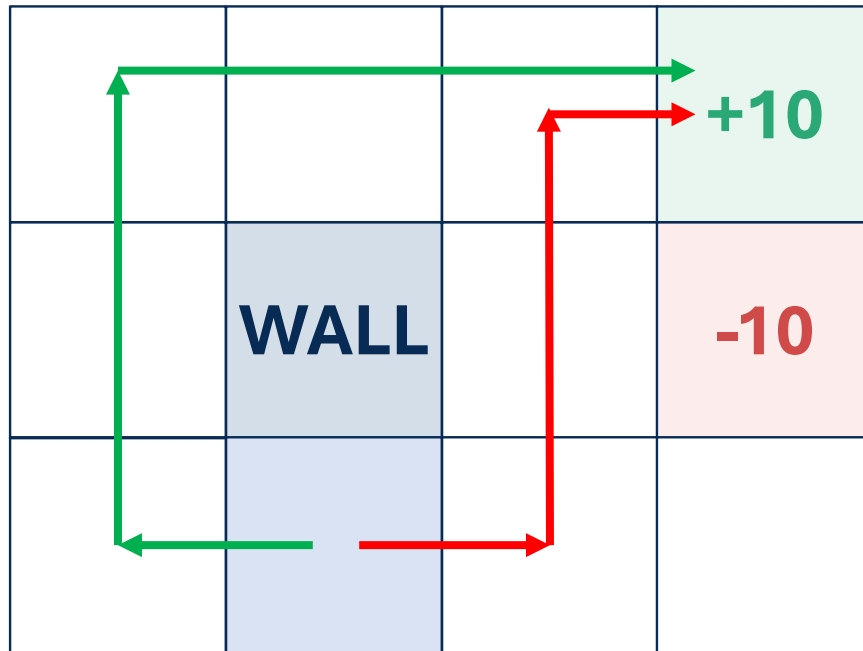
Key idea

Policy tells the agent what to do again after each transition.

The same policy can produce different trajectories because the environment is stochastic.

Immediate reward is not enough

A good policy may accept a smaller reward now to improve future outcomes.



Recall every move costs 1

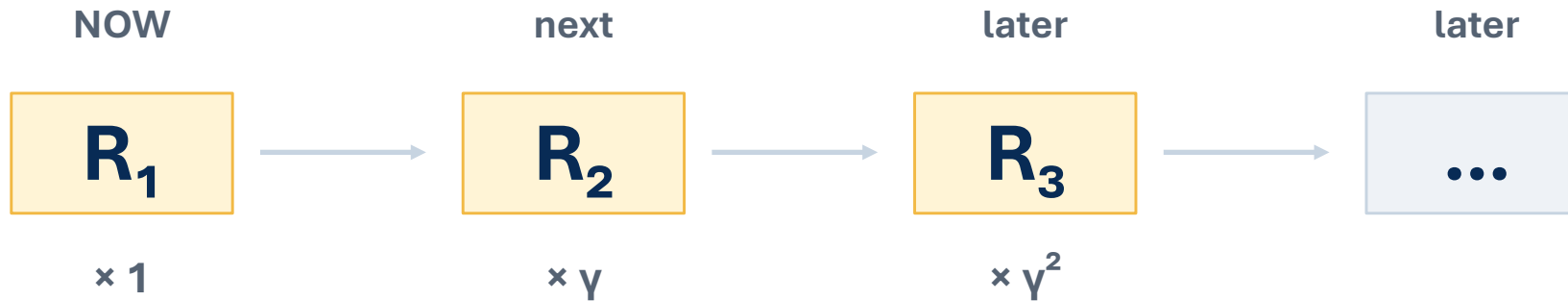
Longer / safer
May look worse now but protect future return.

Shorter / riskier
May have lower immediate cost but passes near bad outcome.

Question: does the best immediate reward prove the best policy?

Return = reward accumulated through time

Return summarizes one future trajectory.



$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

Return is not one reward. It is the reward stream from now into the future.

Discount factor: how much does later count?

γ controls how strongly future reward influences current decisions.

$$0 \leq \gamma \leq 1$$

NOW



1 step



γ

2 steps



γ^2

3 steps



γ^3

Small γ
future rewards fade quickly
agent becomes more short-term

Large γ
future rewards retain influence
agent considers longer consequences

γ is a design choice about valuing future consequences.

γ is not a transition probability

Same numeric range. Different meaning.

$$P(s' | s, a) = 0.8$$

- 80% likelihood of a modeled outcome
- belongs to the transition model
- describes environment uncertainty

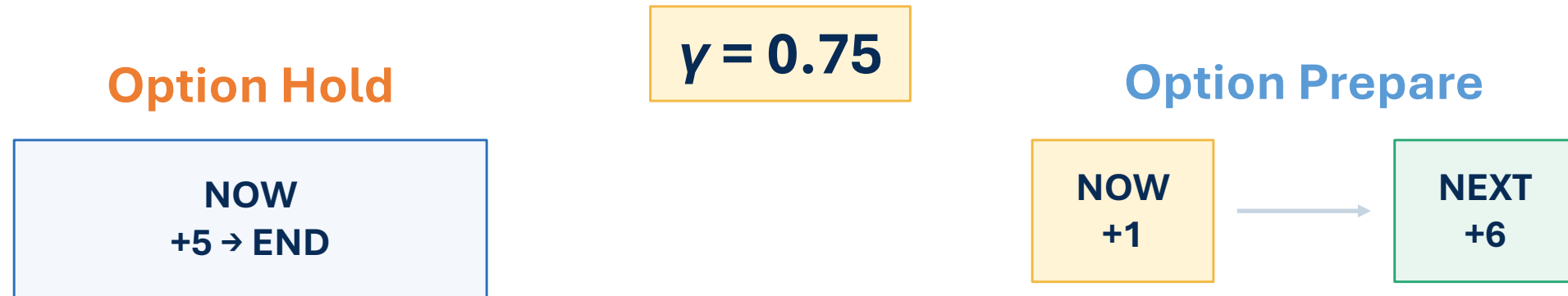
$$\gamma = 0.8$$

- future reward keeps 80% weight one step later
- belongs to the return calculation
- describes how much future reward matters

Not every number between 0 and 1 is a probability.

Visual return problem

A smaller immediate reward can still produce a better long-term decision.



Which action should we choose?

$$G_t = R_{t+1} + \gamma R_{t+2}$$

$$G_{Hold} = 5$$

$$G_{Prepare} = 1 + 0.75(6) = 5.5$$

Why did the preferred action change?
Because future reward can outweigh immediate reward.

Same rewards, different γ , different preference

The environment did not change. The future weight changed.

Option Hold



Option Prepare



$$\gamma = 0$$

Hold: 5

$$\text{Prepare: } 1 + 0(6) = 1.0$$

Preferred: Hold

$$\gamma = 0.5$$

Hold: 5

$$\text{Prepare: } 1 + 0.5(6) = 4.0$$

Preferred: Hold

$$\gamma = 0.9$$

Hold: 5

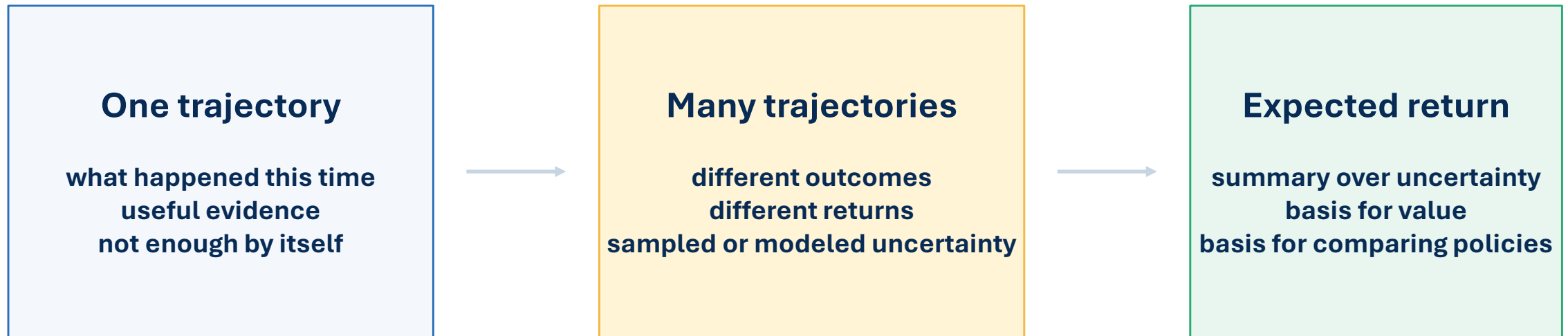
$$\text{Prepare: } 1 + 0.9(6) = 6.4$$

Preferred: Prepare

Discounting changes how strongly future consequences influence behavior.

Policy quality is expected cumulative reward

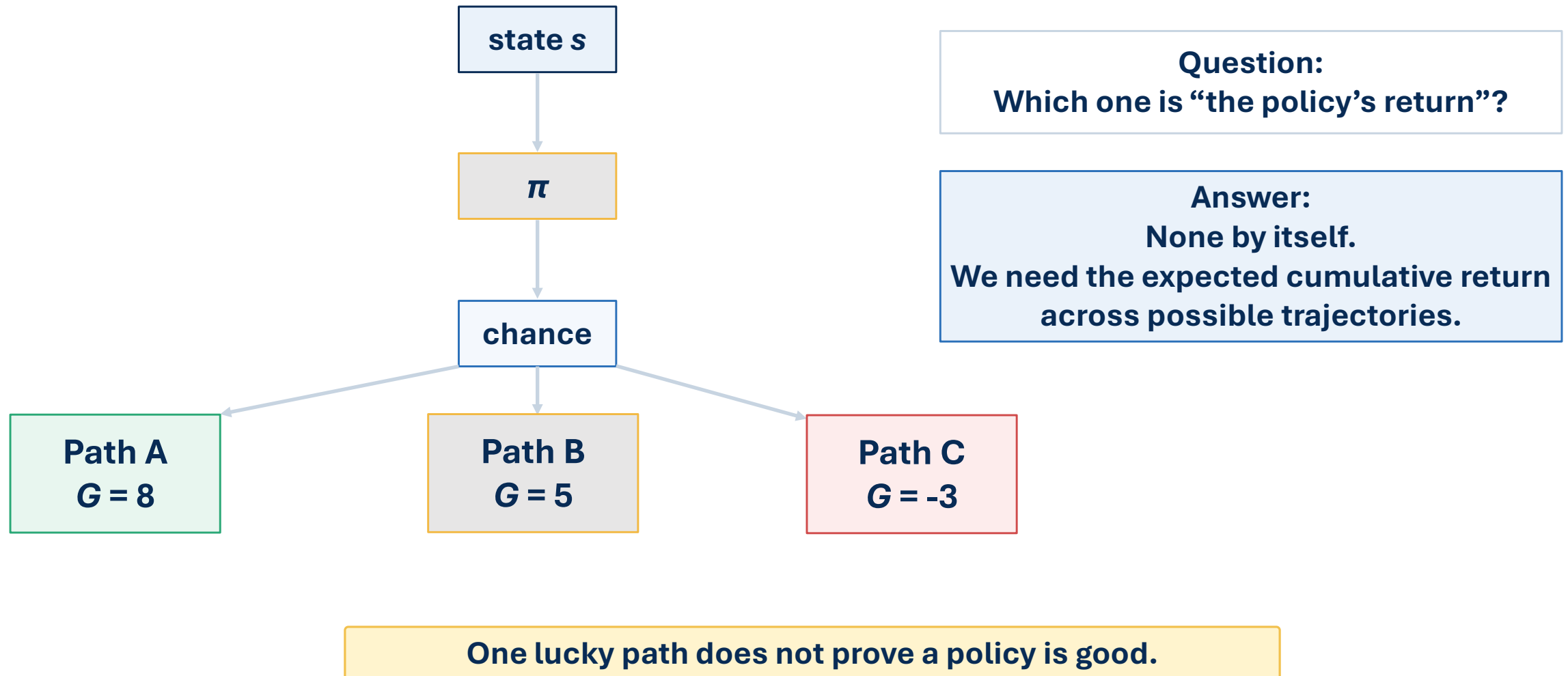
We judge behavior over possible outcomes, not one isolated run.



A good policy is good in expectation over time—not just in the best observed case.

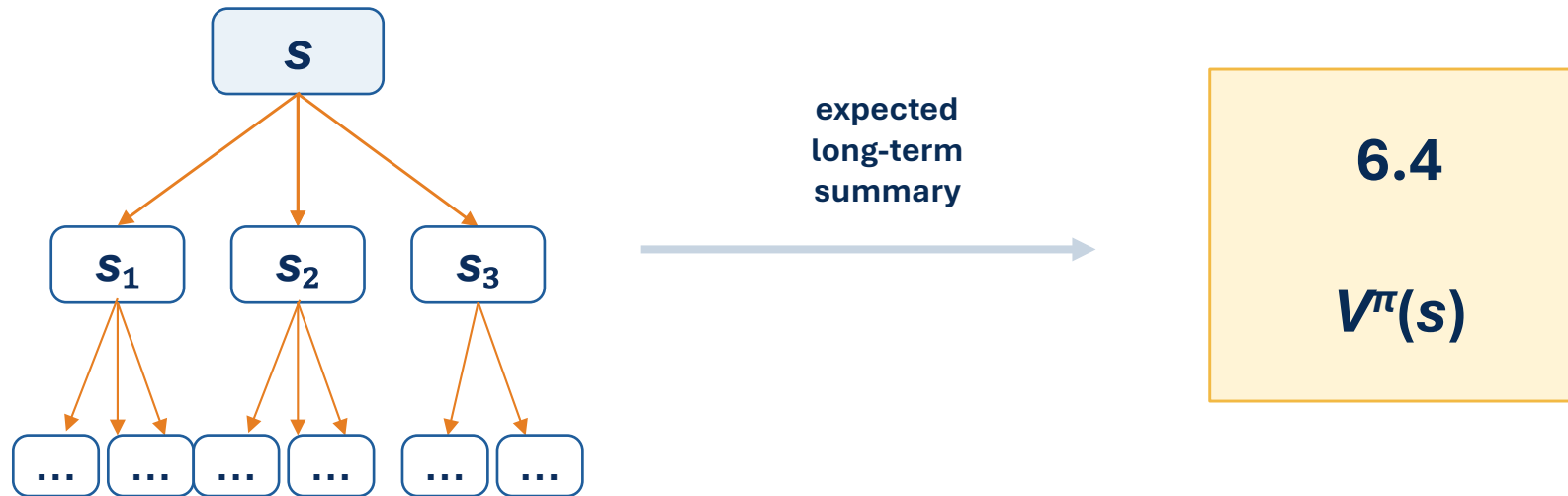
A stochastic world can produce different returns

Same policy, same start state, different trajectories.



Value compresses a whole future tree into one number

Value is an expected long-term summary, not an immediate reward.



A value number stands in for many possible future trajectories.

State value: $V^\pi(s)$

How good is it to start in this state and then follow policy π ?

4.5 →	5.6 →	8.7 →	+10
3.6 ↑	WALL	6.5 ↑	-10
2.8 ↑	3.7 →	4.8 ↑	2.2 ←

State values under policy π

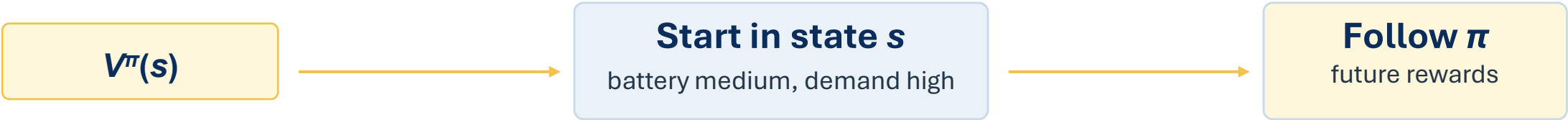
$V^\pi(s)$

Definition

Expected cumulative discounted return starting in state s and then following policy π .

Interpretation warning

These numbers are not immediate rewards. They summarize expected long-term outcomes.



Same state, different policy, different value

Value belongs to a state under a particular policy.

Same physical state
Different policies assume different future behavior.

4.5 →	5.6 →	8.7 →	+10
3.6 ↑	WALL	6.5 ↑	-10
2.8 ↑	3.7 →	4.8 ↑	2.2 ←

Policy π_1

3.0 →	2.6 →	1.7 →	+10
2.2 ↑	WALL	-1.0 ↑	-10
1.5 →	0.8 →	1.2 →	-0.8 ↑

Policy π_2

$$V^{\pi_1}(s) = 3.7$$

This value assumes the agent follows π_1 afterward.

$$V^{\pi_2}(s) = 0.8$$

This value assumes the agent follows π_2 afterward.

The physical state is the same; the behavior being evaluated is different.

Action value: $Q^\pi(s, a)$

How good is it to first take action a here, then follow policy π ?

			+10
	WALL		-10

Zoom into one state-action choice

	5.8	
3.1	S	7.2
	4.0	

Directional Q-values for one state

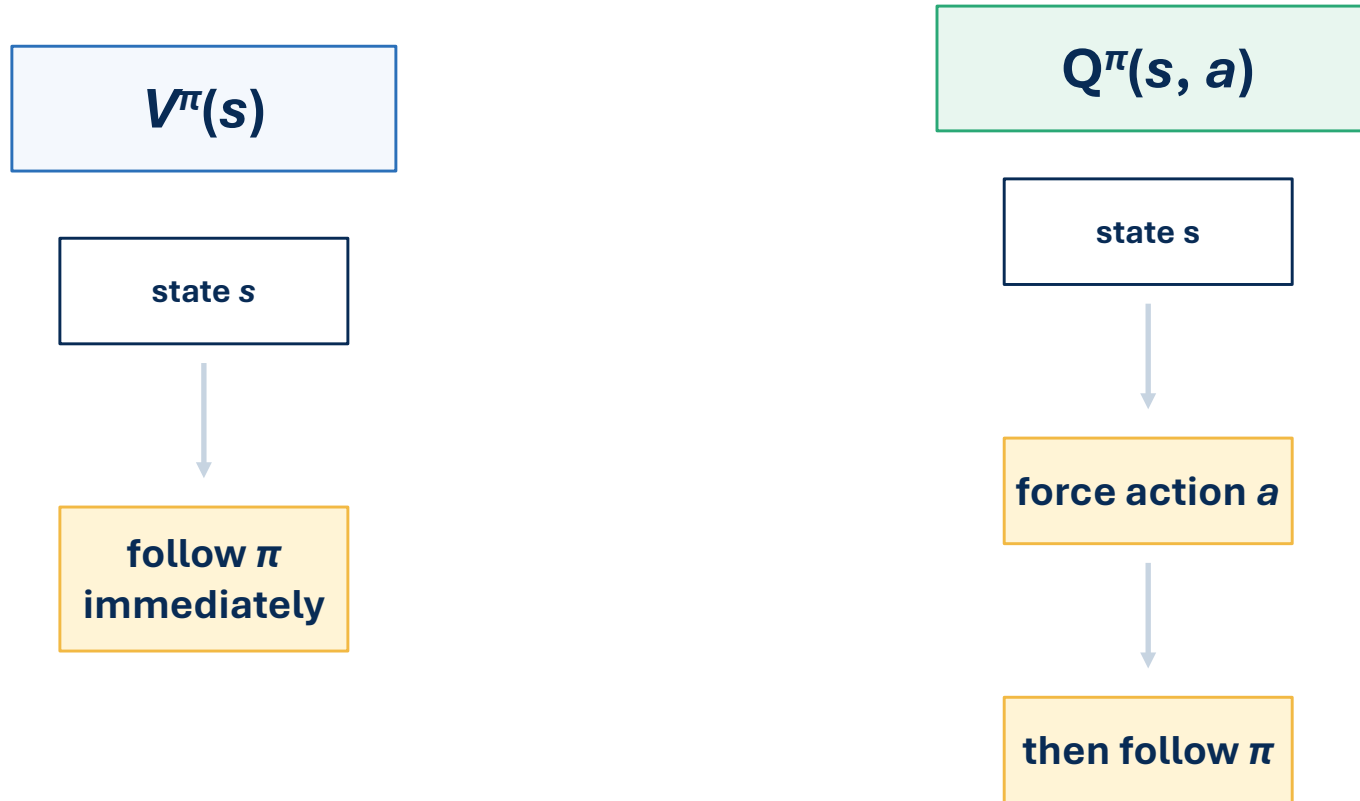
$$Q^\pi(s, a)$$

Expected cumulative return if the first action is a , then the agent follows policy π .

Q asks about doing something there; V asks about being there.

V versus Q: same future, different starting condition

The distinction is about what is fixed at the first step.



Memory cue: V = value of being there. Q = value of doing something there.

A greedy choice reads the Q-values

Choose the action with the highest estimated long-term value.

	5.8	
3.1	s	7.2
	4.0	

Q-values at state s

$$\arg \max_a Q(s, a) = \text{East}$$

Important

7.2 is not the immediate reward.
It is an estimated long-term return.

Greedy does not mean shortsighted
Here, greedy means choosing the largest
available action-value estimate.

What makes a policy optimal?

Optimality is about expected cumulative return.

π^*
Optimal policy

A policy that maximizes expected cumulative return from the states of interest.

**Not necessarily
largest immediate reward**

**Not merely
the best observed lucky run**

**Not required
to be stochastic**

Optimal means best expected long-term behavior—not best immediate-looking move.

Reward → return → value → policy quality

These quantities answer different questions.



Do not collapse these into one idea. They are a chain.

Practice: diagnose the quantity

Identify whether each statement describes Reward, Return, V^π , or Q^π .

1 The robot receives -3 after a failed maneuver.

Reward

2 The discounted sum along this trajectory is 6.8.

Return

3 Starting here and following π has expected return 5.4.

V^π

4 First taking East here, then following π , has expected return 7.1.

Q^π

Partner check: explain why each answer fits.

Practice: Gridworld interpretation sprint

Use the visual representation, not memorized definitions.

4.5 →	5.6 →	8.7 →	+10
3.6 ↑	WALL	6.5 ↑	-10
2.8 ↑	3.7 →	4.8 ↑	2.2 ←

Answer aloud or on paper

What does one arrow represent?

What does one number represent?

Is the number an immediate reward?

If the arrow changes, must the value remain the same?

If Q-values are supplied, which action is greedy?

Goal: connect the visual to the concept before we compute values next lesson.

Practice: long-term decision

Calculate return, then explain the decision

Action A

Reward +4 now
and terminate.

$G = \underline{\hspace{2cm}}$

Action B

Reward +1 now
and +5 next step.

$\gamma = 0.8$

$G = \underline{\hspace{2cm}}$

Explain

- Which return is larger?
- Why is immediate reward misleading?

Synthesis

Rebuild the conceptual chain from memory.

A policy tells us

what action to choose in each state

Return combines

rewards over time

γ

controls how much future reward matters

$V^\pi(s)$ means

expected return from a state under π

$Q^\pi(s,a)$ differs because

it first forces an action

An optimal policy maximizes

expected cumulative return

Exit ticket

Apply the vocabulary to one compact scenario.

Scenario

A controller can either transmit now for +3, or charge now for 0 and transmit next step for +5. Let $\gamma=0.8$.

Answer

1. Return for transmit now
2. Return for charge first
3. Better action by return

Explain

In one sentence: why can the optimal policy reject the largest immediate reward?

Preview L13: We now know what values mean. Next: where do the numbers come from?